

Silverline House — 25-Storey Residential HRB

Geotechnical Ground Investigation Report

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1 Scope and Objectives

This document forms part of the Gateway 2 application for Building Control Approval submitted to the Building Safety Regulator under the Building Safety Act 2022. The information presented herein has been prepared in accordance with the CLC Guidance Suite for new Higher-Risk Buildings and the requirements set out in Guidance Note 04 (Application Information Schedule) and Guidance Note 06 (Document Management and Submission). All design work described in this document has been carried out by suitably qualified engineers and specialists with relevant experience in higher-risk residential buildings. The scope of this document covers the primary structural system, including the reinforced concrete frame, post-tensioned flat slab floors, core walls, and the ground-bearing and piled foundation system. The document does not cover secondary structures such as facade support brackets (addressed separately in the Facade Interface and Support Design document), internal partition framing, or fit-out works. All dimensions and coordinates are in millimetres unless otherwise stated. Grid references correspond to the structural grid shown on drawing SLH-STR-001 Rev A.

The design has been developed in accordance with the Building Regulations 2010 and the relevant Approved Documents. Where European standards (Eurocodes) are adopted, the applicable UK National Annexes have been used. Design standards and codes of practice referenced in this document include, but are not limited to, BS EN 1990 (Basis of structural design), BS EN 1991 (Actions on structures), BS EN 1992 (Concrete structures), BS EN 1993 (Steel structures), and BS EN 1997 (Geotechnical design). Where proprietary systems or products are specified, the manufacturer's technical data and third-party test evidence are referenced in the relevant sections.

2 Ground Conditions

A site investigation was carried out in accordance with BS EN 1997-2 and BS 5930. The investigation comprised twelve rotary core boreholes to a maximum depth of 30 m below existing ground level, together with in-situ standard penetration tests at 1.5 m intervals. Laboratory testing was carried out on selected samples to determine soil classification, shear strength parameters, consolidation characteristics, and chemical contamination. The ground conditions comprise approximately 3 m of made ground overlying Terrace Gravels (3–9 m depth), London Clay (9–22 m), and Lambeth Group deposits below 22 m. Groundwater was encountered at 2.5 m depth in the gravels. The design groundwater level for structural purposes is taken as 0.5 m below existing ground level. The design has been developed in accordance with the Building Regulations 2010 and the relevant Approved Documents. Where European standards (Eurocodes) are adopted, the applicable UK National Annexes have been used. Design standards and codes of practice referenced in this document include, but are not limited to, BS EN 1990 (Basis of structural design), BS EN 1991 (Actions on structures), BS EN 1992 (Concrete structures), BS EN 1993 (Steel structures), and BS EN 1997 (Geotechnical design). Where proprietary systems or products are specified, the manufacturer's technical data and third-party test evidence are referenced in the relevant sections.

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3 Foundation Recommendations

The foundation system consists of 600 mm diameter continuous flight auger (CFA) piles installed to a minimum depth of 18 m below existing ground level, penetrating at least 3 m into the Lambeth Group. Pile design has been carried out in accordance with BS EN 1997-1 using a combination of ground model analysis and empirical correlations from the site investigation data. The working load per pile is 2,800 kN in compression and 800 kN in tension. Settlement analysis indicates maximum pile head settlements of 8 mm under working loads, with differential settlement between adjacent piles not exceeding 5 mm. Pile load testing in accordance with BS EN ISO 22477-1 is required during construction, with at least two maintained load tests to 150% of the working load prior to commencing pile cap construction. Settlement analysis confirms adequate performance under design loads.

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